

Blur Detection Batch Processing System

Non-Technical / Quality Inspector User Manual

Windows Docker Edition

This guide is for daily operators and **does not require programming knowledge**. The system checks photos, organizes the results, and shows the images that need human review. Follow the sections in order.

I. Before You Start

1. Prepare the computer and photos

- Make sure the Windows computer is switched on and working normally.
- If the photos are on a portable hard drive, USB drive, or portable SSD, connect it and confirm that it opens in File Explorer.
- Open the program folder and confirm that you can see `blur_detection.bat`.
- Docker Desktop must be installed and prepared by IT. If this computer has never run the system, contact IT first.

Operators do not install anything: For normal work, you only double-click the start file, select a photo folder, and use the web page. Do not change program files, model files, or Docker settings.

2. Check the image names — this is essential

Every filename must use `SerialNumber-Position`, with one hyphen between the serial number and position.

Example	Correct?	Reason
<code>1359-3.jpg</code>	Correct	Serial number 1359, position 3.
<code>4582A-9.bmp</code>	Correct	Serial number 4582A, position 9.
<code>image_3.jpg</code>	Incorrect	It does not follow the required hyphen format.
<code>1359_pos3.png</code>	Incorrect	Extra text appears before the position.

Each product normally has images for positions **1, 3, 5, 7, and 9**. Incorrect names or missing positions will be sent for review.

Most important: A black command window appears after startup. Keep it open while processing and reviewing. Do not press a key while it is waiting. Closing it or pressing a key at the waiting screen stops the system and makes the web page unavailable.

II. Start the System and Process Photos

- 1 Start the program**
Open the program folder and double-click `blur_detection.bat`. A black window will appear.
- 2 Wait for the system**
If Docker Desktop is not running, the program opens it automatically. This normally takes 20–45 seconds. Do not close any window.
- 3 Tell the system where the photos are**
When the black window asks `Is your data drive external or internal?`, press `E` for a portable drive, USB drive, or portable SSD. Press `I` for a drive inside the computer.
- 4 Select the photo folder**
Select the folder containing all photos, then click OK. Select the folder—not one individual image.

Where will the results be saved?

Photo location	Example photo folder	Result folder
External drive (E)	<code>E:\ProjectA\Photos</code>	<code>E:\pipeline_outputs\ProjectA\Photos</code>
Internal drive (I)	<code>D:\ProjectA\Photos</code>	<code>D:\ProjectA\Photos\pipeline_outputs</code>

The result folder is created automatically.

- 5 Wait for the web page**
The operating page opens automatically. If it does not, open Chrome or Edge, enter `http://localhost:8501` in the address bar, and press Enter.
- 6 Start processing**
Keep the language set to `EN`. On “Batch Process,” confirm that Input is `/data/input`, Output is `/data/output`, and the device is `cuda`. Click “Run Batch Pipeline Process.”

How do I know it is finished? The page displays “Batch Processing Execution Completed Successfully!” together with totals and processing time. Until then, do not disconnect the data drive or close the browser or black window.

Do not change the two path boxes: `/data/input` and `/data/output` are automatic system locations, not Windows folder names.

III. Review and Annotate Images

After batch processing, select “Review & Annotate” at the top of the page. The prediction and image locations are normally filled in automatically.

Review in this order

- **Choose records:** Use the status filter to show Review, FLC, or NG records. If unsure, leave all statuses selected.
- **Enlarge an image:** Hold the mouse pointer over an image. Move it away to return to normal.
- **Check every position:** If the system is wrong, click **O**, **F**, **SN**, **N**, or **NG** beside the image.
- **Confirm the product:** Use “Need FLC,” “Don’t Need FLC,” or “NG (Fail Directly).”
- **Save and continue:** Click “Next” or press . Use “Previous” or  to go back.

1. What do the five labels mean?

Label	Meaning	Use it when
O	OK / Normal	The target position is normal, with no clear offset or distance problem.
F	Far	The target is clearly too far away, severely offset, or incomplete.
SN	Slightly Near	The target is slightly too near, but its boundary can still be seen.
N	Near	The target is much too near, badly blurred or enlarged, or an important area is missing.
NG	Abnormal / Reject	An image is missing, processing failed, or there is an unacceptable serious problem.

2. When does the system automatically require FLC?

FLC means that further abnormality review is required. The system requires FLC if any of these occur across positions 1, 3, 5, 7, and 9:

- At least one position is **N**;
- At least one position is **NG**;
- Three or more positions are **SN**.

Operator tip: The three product-level buttons set the whole product status. If you change a position afterward, the system calculates the status again. Check all five positions before giving the final product status.

IV. View Results and Finish Work

1. Which files should an operator open?

Open the result folder described earlier, then open `dataset_output`. For normal work, use the Excel files ending in `.xlsx`.

File	Purpose
<code>consolidated_batch_predictions.xlsx</code>	System results for every product.
<code>predictions_filtered.xlsx</code>	Only products that need human checking or FLC review.
<code>human_annotated_predictions.xlsx</code>	Final results saved during human review. This is the main file after inspection is complete.
<code>raw_predictions.xlsx</code>	Detailed results for individual images; normally used only for troubleshooting.

Do not change system files: Files ending in `.json` are used by the system. Do not delete, rename, or edit them. The `processed_images` folder contains the processed pictures.

2. Finish work safely

1. Confirm that processing is finished and review changes are saved.
2. Return to the black window and press any key.
3. Wait for the system to stop. Close the window only when it says it is safe.
4. If using a portable drive, safely eject it only after these steps.

3. What should I do if something goes wrong?

What you see	What the operator can do
The web page does not open	Keep the black window open. Start Docker Desktop, wait until it is ready, then run the start file again.
The browser did not open automatically	Enter <code>http://localhost:8501</code> in Chrome or Edge.
The page shows <code>cpu</code>	Stop the program and contact IT. Normal operation should show <code>cuda</code> .
The photo folder cannot be found	Check that the portable drive is connected, then restart and select the folder again.
Many products show missing images	Check filenames and confirm positions 1, 3, 5, 7, and 9 exist for each serial number.
The web page suddenly stops working	Check whether the black window was closed. Restart once; do not start two copies.

Still not working? Do not modify the program. Photograph the black window and web-page error, then send the photos, selected folder location, and time of the problem to IT.